

Word Problems & Applications

Answer Key — Worked Solutions

MTH1W · Homework — Answer Key

Worked solutions with method lines. Curriculum: MTH1W (Grade 9 de-streamed math).

1. **[L1 · Translate]** $n - 7$.
2. **[L1 · Number]** $n + 15 = 40$, so $n = 25$.
3. **[L1 · Money]** $3p = 27$, so $p = 9$ pencils.
4. **[L1 · Percent]** $0.10 \times 250 = \$25$.
5. **[L1 · Distance]** $d = 50 \times 4 = 200$ km.
6. **[L1 · Geometry]** $4s = 48$, so $s = 12$ cm.
7. **[L2 · Translate & solve]** $3n - 8 = 25 \rightarrow 3n = 33 \rightarrow n = 11$.
8. **[L2 · Consecutive integers]** $n + (n+1) = 55 \rightarrow 2n + 1 = 55 \rightarrow n = 27$.
The integers are 27 and 28.
9. **[L2 · Wages]** $60 + 45h = 240 \rightarrow 45h = 180 \rightarrow h = 4$ hours.
10. **[L2 · Mixture]** Juice stays $0.60 \times 2 = 1.2$ L.
 $1.2 = 0.40(2 + w) \rightarrow 1.2 = 0.8 + 0.4w \rightarrow w = 1$ L.
11. **[L2 · Percent (work back)]** $0.80p = 72 \rightarrow p = \90 .
12. **[L2 · Angles]** $x + (x + 40) = 180 \rightarrow 2x = 140 \rightarrow x = 70$.
The angles are 70° and 110° .
13. **[L3 · Equation, both sides]** $5n - 3 = 2n + 12 \rightarrow 3n = 15 \rightarrow n = 5$.
14. **[L3 · Consecutive even]** $n + (n+2) + (n+4) = 78 \rightarrow 3n + 6 = 78 \rightarrow n = 24$.
Integers 24, 26, 28; the largest is 28.
15. **[L3 · Coins]** Let n = nickels: $0.05n + 0.10(20 - n) = 1.65 \rightarrow -0.05n + 2 = 1.65 \rightarrow n = 7$.
So 7 nickels and 13 dimes.
16. **[L3 · Blend cost]** Let x = kg of \$5 tea: $5x + 9(12 - x) = 6(12) = 72 \rightarrow -4x + 108 = 72 \rightarrow x = 9$.
So 9 kg of \$5 tea and 3 kg of \$9 tea.
17. **[L3 · Closing distance]** Combined rate 180 km/h; $180t = 540 \rightarrow t = 3$ hours.
18. **[L3 · Dimensions]** Let w = width, length = $2w + 4$: $2(w + 2w + 4) = 56 \rightarrow 6w + 8 = 56 \rightarrow w = 8$.
Width 8 cm, length 20 cm.